

Lymantria dispar dispar (Spongy Moth)



Department of Conservation and Natural Resources	
Recreation	>
Education	>
Conservation	>
Climate Change	>
Geology	>
Sustainable Practices	>
Forests and Trees	>
State Forest Management	
Insects and Diseases	
Prescribed Fire	
Natural Gas Drilling Impact	
Managing Your Woods	
Fall Foliage Reports	
Forest Types	
Wild Plants	>
Wildlife and Biodiversity	>
Water	>
Programs and Services	>
Find an Event	
Make a Reservation	
Interactive Maps	
Newsroom	
Contact Us	
DCNR Search	

Spongy moth (*Lymantria dispar dispar*) — formerly known as gypsy moth — is a serious forest pest and is responsible for killing millions of oak and other species of trees across the state.

Although oak species are preferred, spongy moth caterpillars feed on hundreds of other tree and shrub species, including:

- Apple
- Alder
- Aspens
- Basswood
- Birches
- Hawthorn
- Hemlock
- Tamarack (larch)
- Pines
- Spruces
- Willows
- Witch hazel

Often confused with other caterpillars, the spongy moth caterpillar has a yellow and black head, a hairy body with five pairs of blue spots, followed by six pairs of red spots.

It usually takes more than one year of defoliation before trees die; however, conifers that are defoliated may be killed after a single season of defoliation.

Although the boom and bust cycles of the spongy moth are less severe than during the past, they still require control during years when their populations are high.

Tips for Homeowners with Spongy Moth Infestations

Survey your property for egg masses in the summer and fall. The spongy moth has one generation per year in Pennsylvania.

Females lay their eggs as light tan egg masses (100–1,500 eggs/mass) on trees, stones, and other substrates during June and July.

Eggs hatch from mid-April to early May the following spring.

Penn State Extension offers a [Guide to Spongy Moth Egg Mass Surveying](#), which includes an instructional video, both of which explain how to survey for spongy moth egg masses.

Mechanical Removal of Spongy Moth Egg Masses and Larvae

Tactics for mechanical removal of spongy moth egg masses can be effective for individual yard trees, but are not effective as a forest-wide control method.

Methods include removal of egg masses before they hatch and removal of unnecessary yard objects where egg masses can be hidden, such as:

- Piles of old wood
- Building materials
- Dead branches, firewood, and other refuse

Egg masses should be scraped into a sealed container or bag and disposed.

Another control tactic is wrapping burlap around the trunks of trees where spongy moth larvae can hide during the day.

The larvae hiding under the burlap are then scraped into a can of soapy water, killing them.

Insecticide Treatments to Control Spongy Moth Caterpillars

Using insecticides to reduce defoliation during high spongy moth densities is an effective option; however, they do not eliminate the spongy moth entirely or shorten the infestation period.

Homeowners must assess the risks and benefits of insecticide use.

Insecticides are not necessary unless the population of spongy moth egg masses or larvae indicate a threat to your trees.

There are several insecticides registered for use for spongy moth; however, only some of them are available for homeowner use.

Always read the label instructions before using an insecticide.

Spongy Moth Spraying for Private Residential Landowners

DCNR conducts an aerial spongy moth suppression program to treat state and federal forest lands.

DCNR does not spray for spongy moths on private lands.

Private landowners or municipalities can conduct their own treatment program by following the steps in the [Guide to Conducting a Private Spongy Moth Suppression Program \(PDF\)](#).

A recent innovation is the use of drones to chemically treat smaller woodlots or residential trees.

[Aerial applicators licensed to work in Pennsylvania \(PDF\)](#). (Attention smart phone users: Be sure to select "Download file in native format" for full document display.)

Treatments begin in the spring (usually May) when 50 percent of the caterpillars are in their second instar, so timing is critical.

If you believe that you have a need for a spongy moth suppression treatment, you should begin making plans during the fall and winter months preceding the year of treatment.

Spongy Moth Infestations in Pennsylvania Forests

The longest-standing effort to manage forest pests on Pennsylvania's forest lands is the Department of Conservation and Natural Resources Bureau of Forestry's spongy moth (previously known as gypsy moth) program.

The spongy moth has been causing significant forest damage in Pennsylvania since the 1970s.

An outbreak occurred from 2013 to 2019 and another outbreak is occurring from 2021 through 2025.

The Bureau of Forestry uses an integrated pest management approach to monitor and treat spongy moth populations to lessen tree mortality and prevent significant defoliation.

The bureau conducts [annual spongy moth egg mass surveys \(PDF\)](#) to monitor spongy moth populations and implements a suppression program when populations exceed threshold levels.

The bureau uses applications of *Bacillus thuringiensis*, subspecies *kurstaki* (Btk) — a natural biological insecticide — and tebufenozide — an insect growth regulator — to control spongy moth populations via aerial application using airplanes and helicopters.

Both insecticides are deemed safe and are commonly used in agriculture.

2025 State and Federal Land Spray Area Map

The [Lymantria dispar dispar \(Spongy Moth\) Spraying Status GIS Map](#) provides information about where the Department of Conservation and Natural Resources will be spraying for spongy moth infestations on state and federal lands in 2025.

2025 Pennsylvania Game Commission Spray Map

The Pennsylvania Game Commission will conduct their own spongy moth suppression program in 2025.

These spray blocks can be found on the [Pennsylvania Game Commission Spongy Moth Spray Blocks](#).

Additional Resources

2025 Spongy Moth Spraying Documents

2025 Spongy Moth Spraying Program Fact Sheet (PDF)

2025 Spongy Moth Spray Program Map (PDF)

BTK Insecticide Fact Sheet (PDF)

Tebufenozide (MIMIC 2LV) Information (PDF)

Tebufenozide (MIMIC 2LV) Guidelines for Use on State Forests (PDF)

2025 Environmental Review

Lymantria dispar dispar Suppression Project Environmental Description 2025 (PDF)

Lymantria dispar dispar Biological Evaluation of Outbreaks (PDF)

U.S. Forest Service Lymantria dispar Digest

Was this page helpful?

☐ Yes

☐ No

Return to top

TOP SERVICES

- Register to Vote
- Find a DMV
- Get a Birth Certificate
- Join the Veterans Registry
- PAYback

PA.GOV

- Careers & Internships
- PennWatch
- Right-to-Know Law